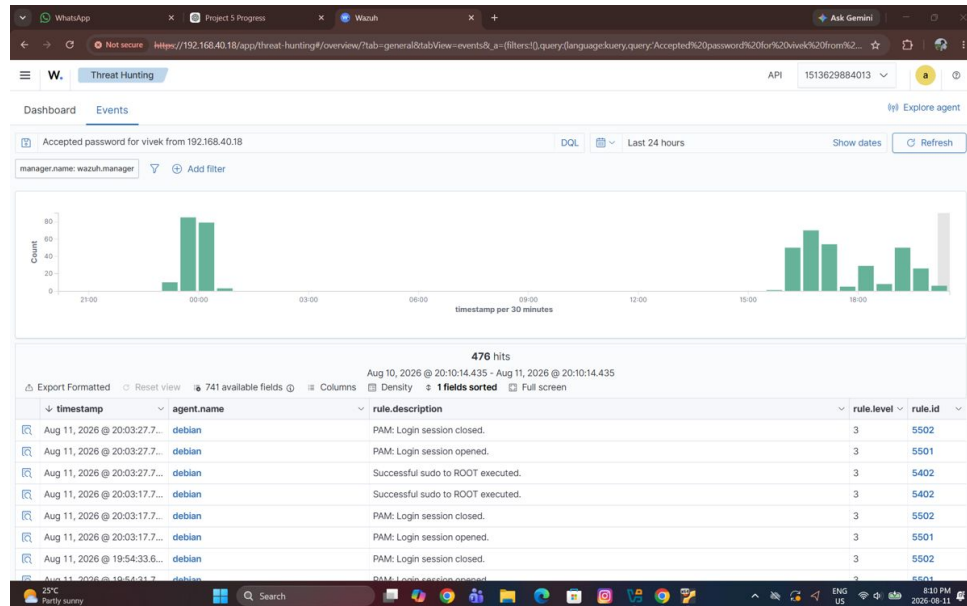


Project 5 - SOC & SIEM

04 - Attack Simulation Evidence

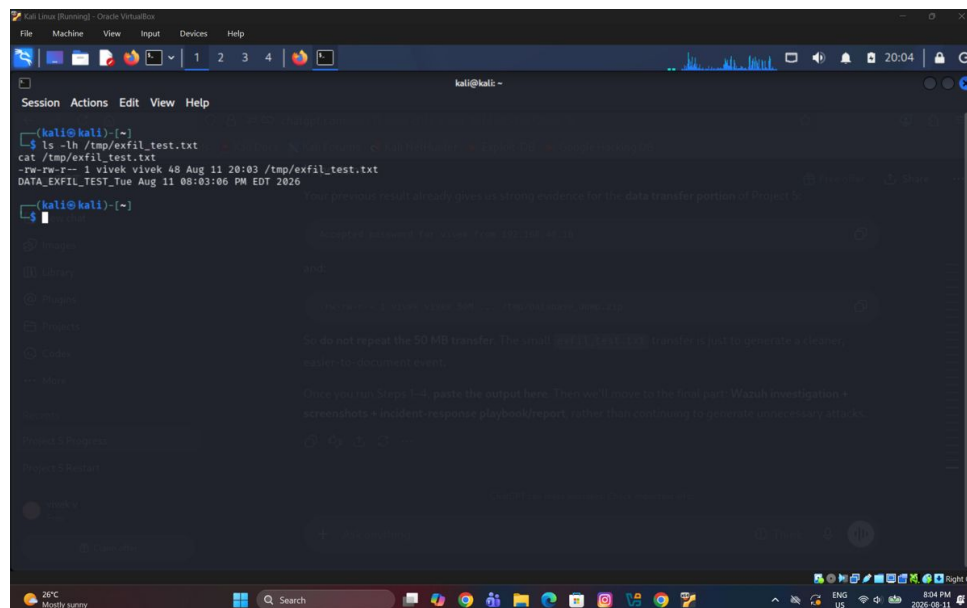
Controlled simulations were performed in the isolated lab to validate the SIEM workflow without affecting production systems. The evidence below covers authentication activity, controlled testing, endpoint investigation, and file-transfer-related validation.

Wazuh Threat Hunting Overview



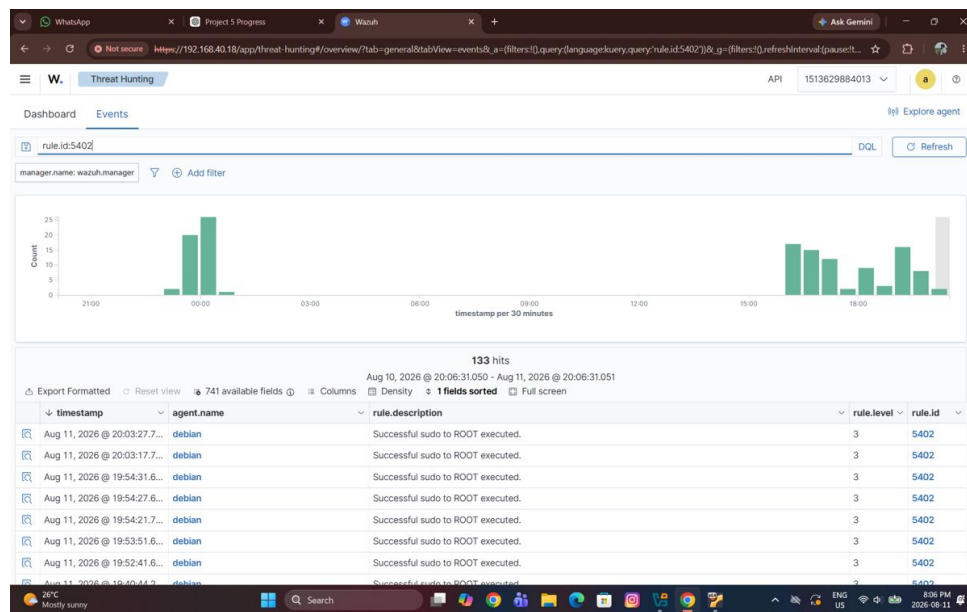
Shows centralized Wazuh event activity used to validate that the controlled lab simulations were generating security telemetry visible to the SIEM.

Kali Linux Controlled Testing Terminal



Shows the Kali Linux terminal used during controlled security testing. Kali acted as the testing host for authentication and other lab validation activities.

Wazuh Event Review During Controlled Test



Shows the Wazuh event view used to review security events generated during the controlled simulation and confirm that the activity was visible for investigation.

Linux/Wazuh Testing Workflow Evidence

```
Aug 11 16:57:10 debian sshd-session[15208]: Failed password for vivek from 192.168.40.59 port 42506 ssh2
Aug 11 16:57:10 debian sshd-session[15205]: Failed password for vivek from 192.168.40.59 port 42478 ssh2
Aug 11 16:57:10 debian sshd-session[15204]: Failed password for vivek from 192.168.40.59 port 42466 ssh2
Aug 11 16:57:10 debian sshd-session[15212]: Failed password for vivek from 192.168.40.59 port 42520 ssh2
vivek@debian:~$ sudo docker exec -it single-node-wazuh.manager-1 bash
bash-5.2# grep -i "sshd" /var/ossec/ruleset/rules/0095-sshd_rules.xml | head
<group name="sshd" >
  <decoded as="sshd" >
    <description>SSH messages grouped.</description>
    <description>sshd: Possible attack on the ssh server (or version gathering).</description>
    <description>sshd: Reverse lookup error (bad ISP or attack).</description>
    <description>sshd: Possible breakin attempt (high number of reverse lookup errors).</description>
    <description>sshd: Timeout while logging in.</description>
    <description>sshd: Possible scan or breakin attempt (high number of login timeouts).</description>
    <description>sshd: insecure connection attempt (scan).</description>
    <description>sshd: OpenSSH challenge-response exploit.</description>
  </group>
bash-5.2# grep -A5 -B2 "Failed password" /var/ossec/ruleset/rules/0095-sshd_rules.xml
<rule id="5760" level="5">
  <if_sid>5700,5716</if_sid>
  <match>Failed password|Failed keyboard|authentication error</match>
  <description>sshd: authentication failed.</description>
  <mitre>
    <id>T1110.001</id>
    <id>T1021.004</id>
  </mitre>
bash-5.2# cat >> /var/ossec/etc/rules/local_rules.xml <<'EOF'
<group name="local,sshd,bruteforce,">
  <rule id="100003" level="10">
    <if_matched_sid>5760</if_matched_sid>
    <frequency>5</frequency>
    <timeframes>60</timeframes>
    <description>Possible SSH brute force attack detected from multiple failed login attempts</description>
    <group>authentication_failures,bruteforce,</group>
  </rule>
</group>
```

Shows terminal-based Linux/Wazuh evidence from the testing workflow, supporting validation of the controlled activity from the endpoint and SIEM sides.